

Question 2: CPU Scheduling

Consider the following information about processes (P1 to P5): (20 Marks)

Process	Arrival Time	CPU Burst-Time	Priority
P1	0	8	3
P2	3	4	2
P3	5	4	1
P4	7	2	1
P5	8	3	0

Draw Gantt Chart, then calculate s, the average waiting time, the average response time, and the average turnaround time of the following scheduling algorithms:

1. First Come First Serve (FCFS) algorithm. (5 Marks)
2. Non-preemptive Shortest Job First (SJF) algorithm. (5 Marks)
3. Non-preemptive SJF with priority algorithm. (5 Marks)
4. Preemptive Shortest Remaining Time First with priority algorithm. (5Marks)

Note: Process priority is only used to discriminate between processes having the same CPU-burst time in the time of taking decision and the lowest integer represents the highest priority.

حل سؤال CPU Scheduling

(Formulas) : القوانين المستخدمة

CT = Completion Time

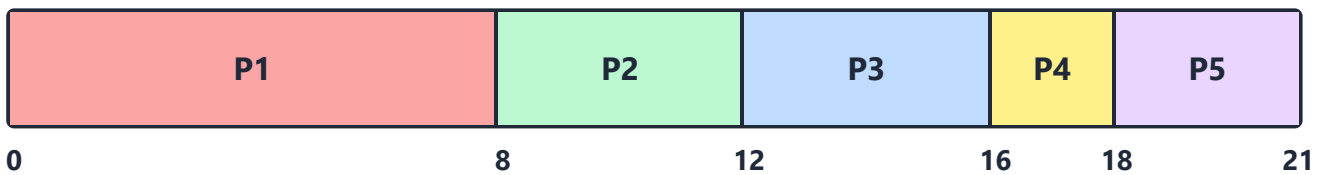
TAT (Turnaround Time) = CT - Arrival Time

WT (Waiting Time) = TAT - Burst Time

RT (Response Time) = First Start Time - Arrival Time

1. First Come First Serve (FCFS)

Gantt Chart



Process	AT	BT	CT	TAT	WT	RT
P1	0	8	8	8	0	0
P2	3	4	12	9	5	5
P3	5	4	16	11	7	7
P4	7	2	18	11	9	9
P5	8	3	21	13	10	10

Average Waiting Time (AWT) = $(0 + 5 + 7 + 9 + 10) / 5 = 31 / 5 = 6.2$

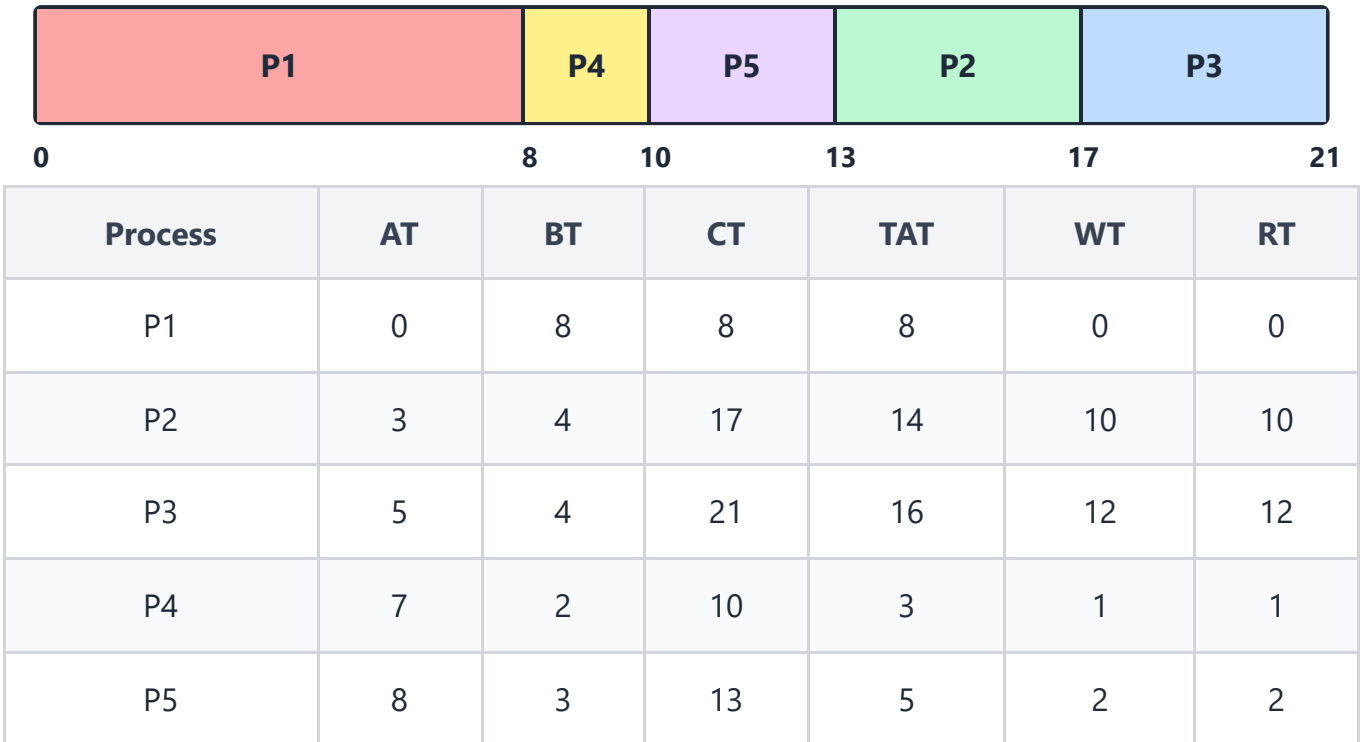
Average Response Time (ART) = $(0 + 5 + 7 + 9 + 10) / 5 = 31 / 5 = 6.2$

Average Turnaround Time (ATAT) = $(8 + 9 + 11 + 11 + 13) / 5 = 52 / 5 = 10.4$

2. Non-preemptive Shortest Job First (SJF)

ملاحظة: لفك التعادل بين العمليات التي لها نفس ال Burst Time (P2 و P3)، تم استخدام وقت الوصول (Arrival Time) بناءً على الخوارزمية القياسية.

Gantt Chart



Average Waiting Time (AWT) = $(0 + 10 + 12 + 1 + 2) / 5 = 25 / 5 = 5.0$

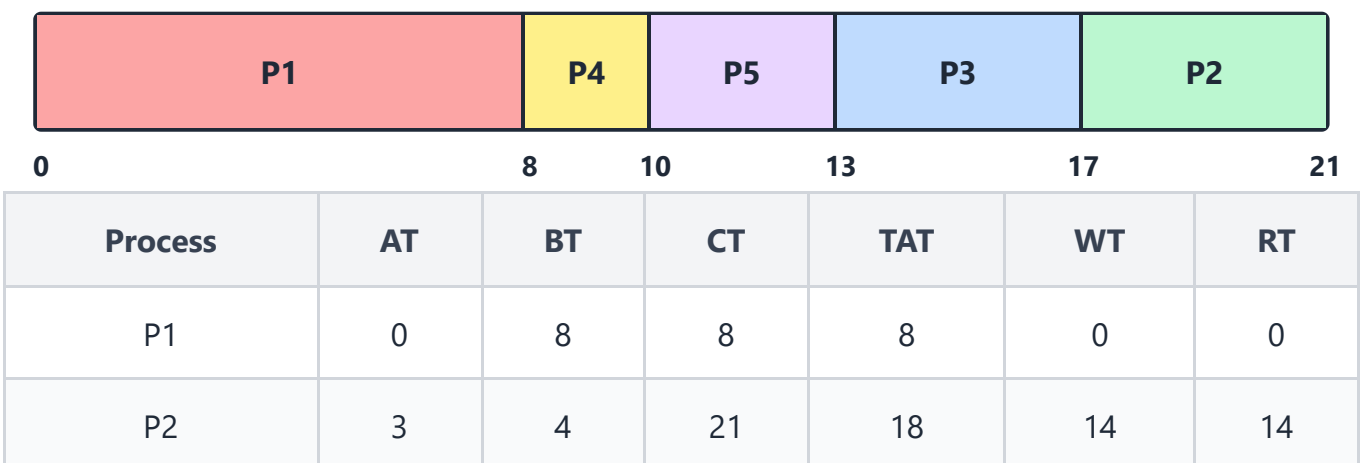
Average Response Time (ART) = $(0 + 10 + 12 + 1 + 2) / 5 = 25 / 5 = 5.0$

Average Turnaround Time (ATAT) = $(8 + 14 + 16 + 3 + 5) / 5 = 46 / 5 = 9.2$

Non-preemptive SJF with priority .3

ملاحظة: لفك التعادل بين العمليات التي لها نفس ال Burst Time (P2 و P3)، تم استخدام الأولوية (Priority) كما ذكر في ملاحظة السؤال.

Gantt Chart



Process	AT	BT	CT	TAT	WT	RT
P3	5	4	17	12	8	8
P4	7	2	10	3	1	1
P5	8	3	13	5	2	2

Average Waiting Time (AWT) = $(0 + 14 + 8 + 1 + 2) / 5 = 25 / 5 = 5.0$

Average Response Time (ART) = $(0 + 14 + 8 + 1 + 2) / 5 = 25 / 5 = 5.0$

Average Turnaround Time (ATAT) = $(8 + 18 + 12 + 3 + 5) / 5 = 46 / 5 = 9.2$

Preemptive Shortest Remaining Time First (SRTF) with .4 priority

ملاحظة: في هذا المسار، لم يحدث تعادل في الوقت المتبقي (Remaining Time) يستدعي استخدام الأولوية كفاصل.

Gantt Chart

P1 P2 P4 P5 P3 P1						
0	3	7	9	12	16	21
Process	AT	BT	CT	TAT	WT	RT
P1	0	8	21	21	13	0
P2	3	4	7	4	0	0
P3	5	4	16	11	7	7
P4	7	2	9	2	0	0
P5	8	3	12	4	1	1

Average Waiting Time (AWT) = $(13 + 0 + 7 + 0 + 1) / 5 = 21 / 5 = 4.2$

Average Response Time (ART) = $(0 + 0 + 7 + 0 + 1) / 5 = 8 / 5 = 1.6$

Average Turnaround Time (ATAT) = $(21 + 4 + 11 + 2 + 4) / 5 = 42 / 5 = 8.4$